

ICT Handover Package: JSS3 Diagnostic CBT System

Date: 2026-07-03
Project: Terminal Examination & SS1 Pathway Placement
Target Audience: ICT/Technical Implementation Team

1. System Overview & Objective

The goal is to deploy a Computer-Based Test (CBT) that serves as both a terminal exam for JSS3 and a diagnostic tool for Senior Secondary (SS1) placement. The system must automatically analyze student performance and recommend one of three academic pathways: **Science, Commercial, or Arts**.

2. Database & Metadata Requirements

Every question in the provided PDF banks must be uploaded with the following metadata tags to enable the diagnostic engine:

Field Name	Description	Example
Subject	The main subject area.	Mathematics, Physics, Economics
Topic	The broad curriculum topic.	Algebra, Demand & Supply, Tawhid
Subtopic	The specific concept being tested.	Quadratic Equations, Price Elasticity, Pillars of Islam
Difficulty	Level of cognitive demand.	Easy, Medium, Tough
Skill Tag	The specific skill being assessed.	Abstract Reasoning, Numerical Fluency, Textual Analysis
Pathway	Which SS1 track this question supports.	Science, Commercial, Arts

3. The Recommendation Engine Logic

The system must use the following logic to generate the "Academic Pathway Recommendation" in the student report:

Science Track: High performance in Mathematics + (Physics OR Chemistry OR Biology).
Commercial Track: High performance in Mathematics + (Economics OR Accounting OR Commerce).
Arts Track: High performance in English Language + (Literature OR Government OR IRS/CRS).

4. Question Bank Inventory

The following PDF files contain the complete question banks. Ensure all options, correct answers, and diagnostic notes are accurately transcribed into the CBT database.

Subject	File Name	Question Count
Mathematics	complete_jss3_math_diagnostic_bank_60q.pdf	60
English Language	english_diagnostic_part_1_4.pdf (4 files)	60
Basic Science	complete_jss3_basic_science_diagnostic_bank_60q.pdf	60
Financial Accounting	ss1_accounting_diagnostic_bank_40q.pdf	40
Economics	ss1_economics_diagnostic_part_1_2.pdf (2 files)	40
Commerce	ss1_commerce_diagnostic_part_1_2.pdf (2 files)	40
Literature-in-English	ss1_literature_diagnostic_part_1_2.pdf (2 files)	40
Government	ss1_government_diagnostic_part_1_2.pdf (2 files)	40
Islamic Religious Studies	ss1_irs_diagnostic_bank_40q.pdf	40

5. Reporting Dashboard Requirements

- **Student View:** Must show a "Strengths & Weaknesses" radar chart and a clear "Recommended Pathway" banner.
- **Teacher View:** Must show "Item Analysis" (which questions were missed most often) to guide remediation.
- **Admin View:** Must show the distribution of students across the three pathways (Science/Commercial/Arts).

6. Security & Integrity

Ensure the CBT platform uses randomized question selection from the banks to prevent malpractice. The "Diagnostic Focus" notes should be hidden from students during the exam and only visible in the post-exam report.